PRACTICAL EXERCISE · WEEK 3

Construction of a Wheeled Inverted Pendulum

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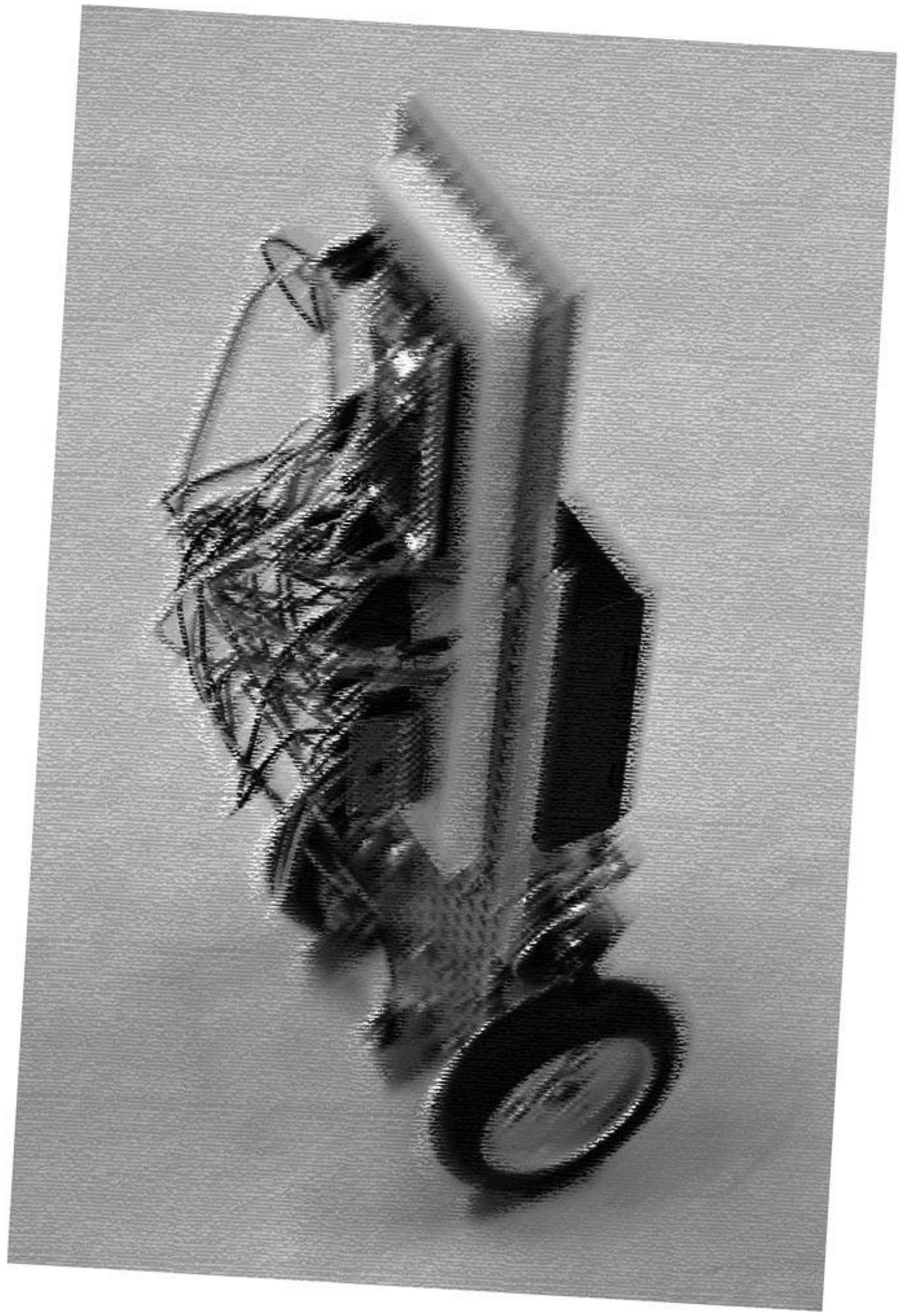
COURSE Department of Precision Engineering · 2026 S1S2

SESSIONS Mondays, 13:00–18:00 · Room 326, Engineering Bldg. 14

V1.2 2026·05·11



Timeline

1

Week 1

Standing

Assembly & bring-up

Angle-only **standing** demo

2

Week 2

Position control

Build a rotary encoder

Position control

Simulation in Simulink

3

Week 3

EXTENSION

Load & disturbance tests

Advanced control (LQR/MPC)

……

4

Week 4

presentation

10-minute presentation

Live demo & Q&A

Final grade · 40%

Week 3 is for extended tasks, or you can continue working on the tasks of Week 1 and Week2

Week 3 - Extension Tasks

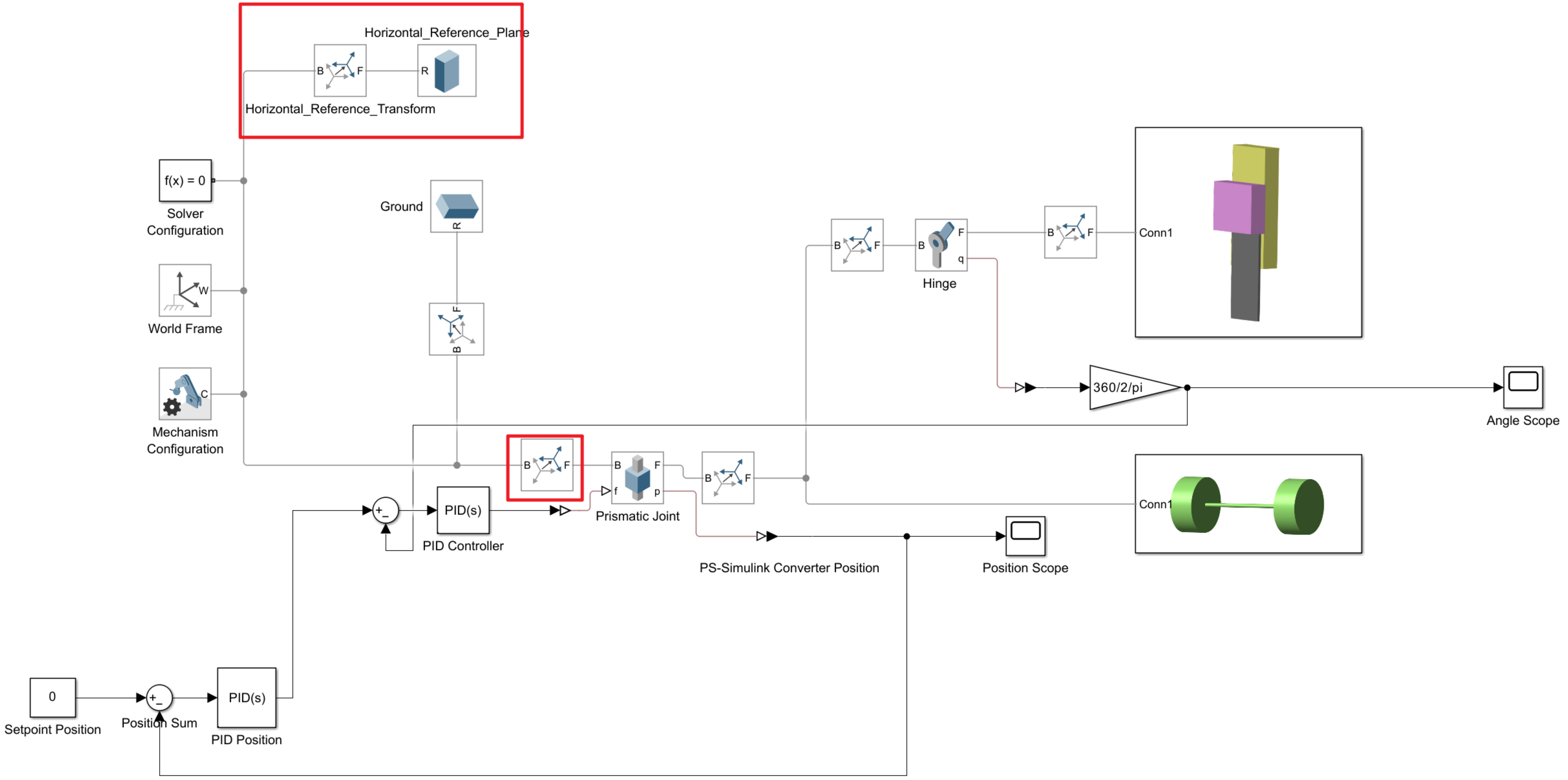
Reference Topics

1. Load Test
2. Enhanced Simulation
3. Advanced Controller
4. Slope Challenge
5. Back-and-Forth Motion
6. Maximum Disturbance

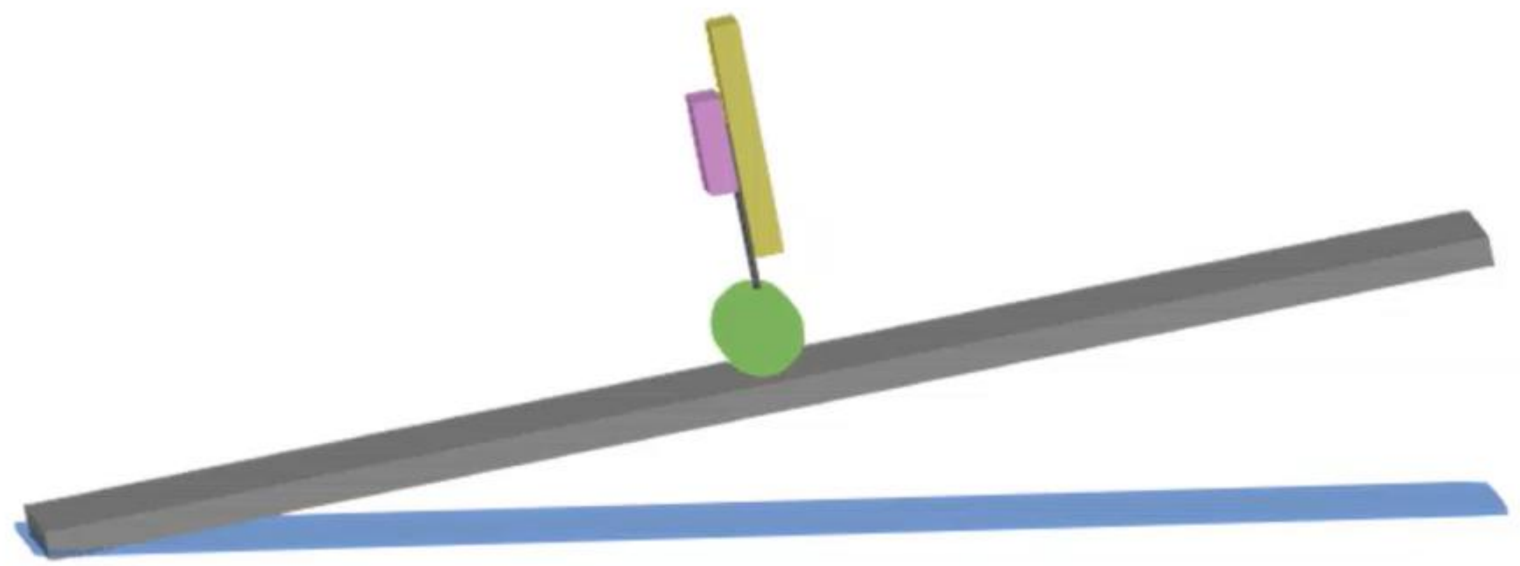
Assign the tasks among yourselves.

Can be done in hardware or simulation.

Simulink Simulation - Slope Challenge



Simulink Simulation - Slope Challenge



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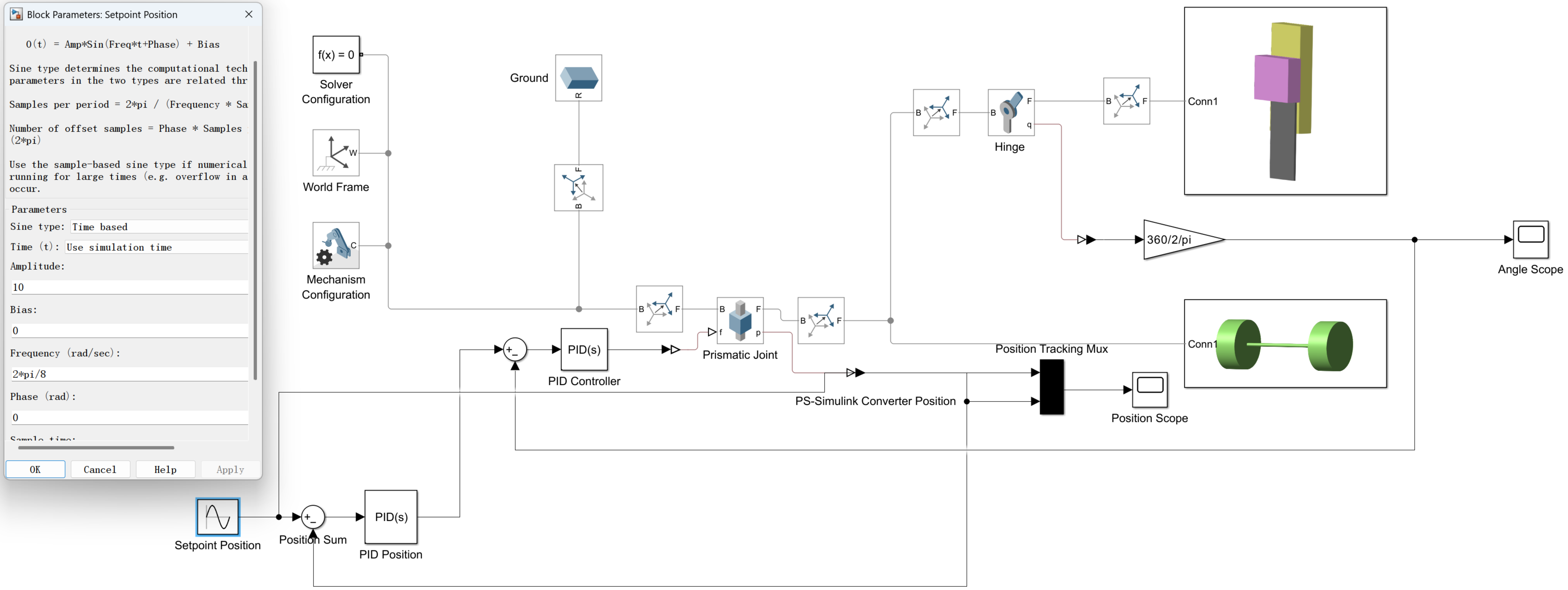
Simulink Simulation - Slope Challenge

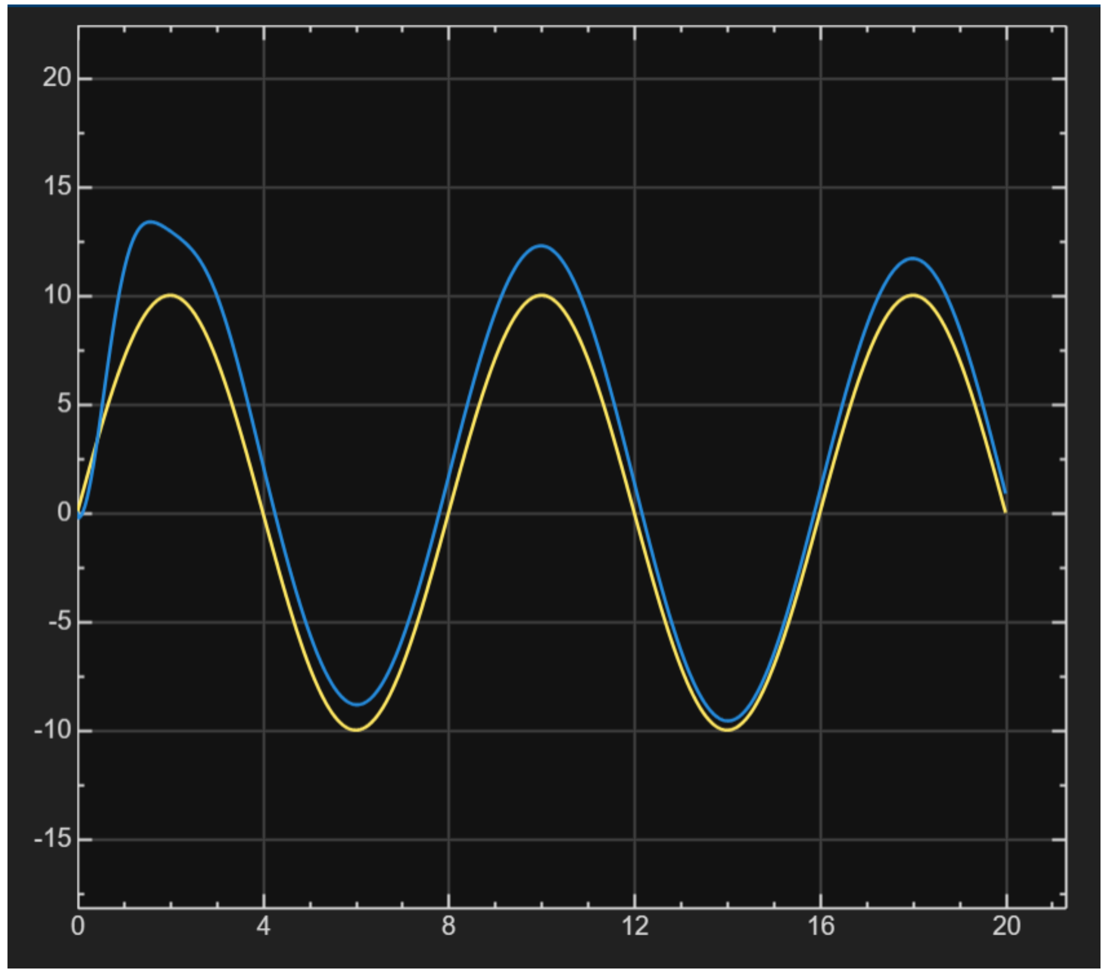
~ 10°

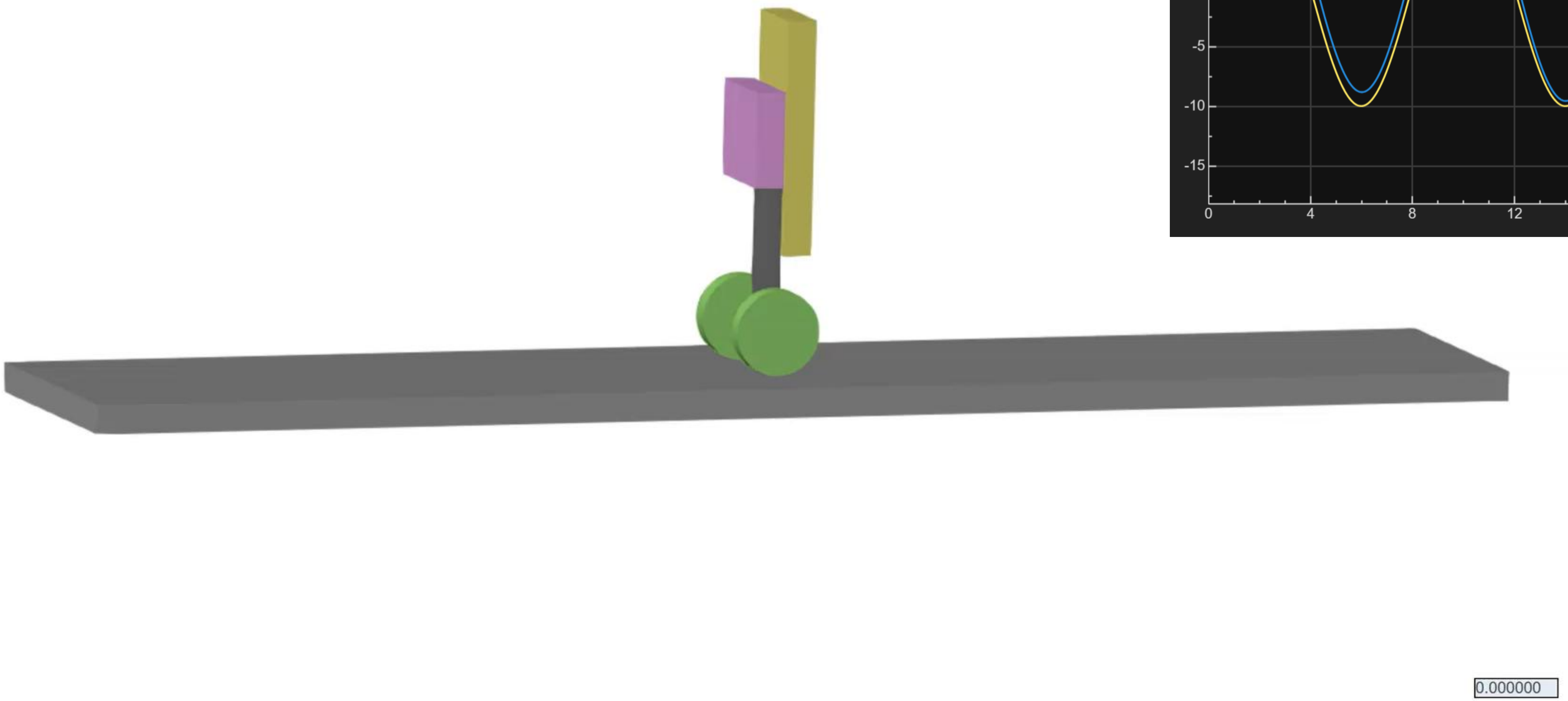
~ 15°

Simulink Simulation - Back-and-Forth Motion



Simulink Simulation - Back-and-Forth Motion





Week 3 Submission

Video 1:

The largest disturbance or load that your real wheeled inverted pendulum can withstand.

Video 2:

The largest disturbance that your simulation model can withstand under comparable conditions

One-page summary (PDF):

explain what maximum disturbance or load you applied, what you improved compared with the previous stage, and, if the system did not work, explain why it did not work.

DEADLINE

21:00

on the day of the class

Today — May 11

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

Each group only needs one submission.

Start from 3:30 pm Next week

WEEK 4 · FINAL SHOWCASE PREVIEW

8 slides - 7 minutes + 3-minute demo + 5-min Q&A

01
Introduction
Project goal, team task, robot configuration. State the problem and why it matters.

02
Method
Modeling, control design, sensor usage, implementation strategy.

03
Experiment
Test setup, conditions, evaluation procedure. How did you measure success?

04
Data
Key plots, tables, numerical results — the evidence behind your verdict.

05
Extended Experiment
Beyond the basic task — load test, parameter study, sim vs. real, alternative controller.

06
Results & Future Work
What worked, what didn't, what to improve next. Honest assessment over hype.

07
What You Learned
Reflect on lessons — technical, experimental, teamwork, debugging.

08
Teamwork & Roles
Who did what — task split, collaboration, individual contributions and shared decisions.

+3
MINUTES
Demo Video or Live Run
A video or run the robot live